

CrisisLens v2.0

災情分類與應變建議系統

Disaster Classification & Response Advisory System

SDG 11 | Sustainable Cities

SDG 13 | Climate Action

Neural Network + MLOps

Group Final Report | June 2026

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Neural Network
Models

6

Disaster
Classes

H3

Hexagonal GIS
Heatmap

RAG

FAISS + Gemini
Advisory

MLOps

Version
Tracking

Problem Statement & SDG Alignment

The Challenge

1. Taiwan faces ~4 typhoons, multiple earthquakes, and flooding events every year
2. Citizens lack a fast, AI-powered tool to classify disasters and receive immediate guidance
3. Emergency managers cannot aggregate scattered citizen reports into a spatial picture in real-time
4. Delayed triage increases harm — the first 30 minutes of disaster response are critical

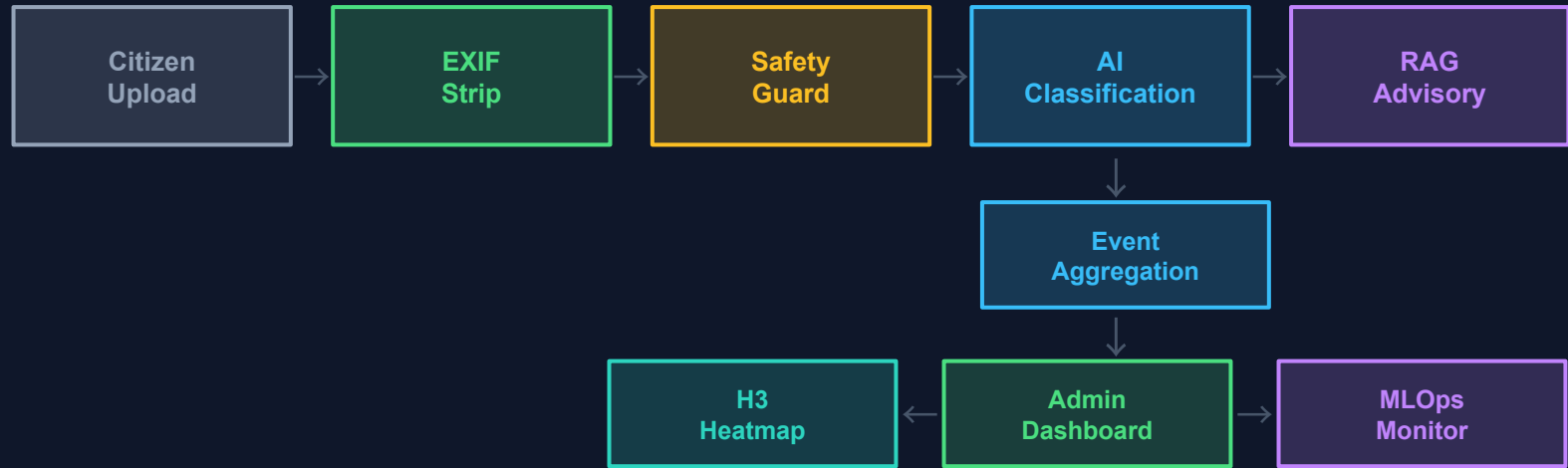
SDG 11 — Sustainable Cities

Enable rapid urban disaster reporting and response coordination, reducing harm in dense communities through AI-powered triage and spatial event aggregation.

SDG 13 — Climate Action

Address the growing frequency of climate-driven disasters — typhoons, floods, and landslides — through AI-assisted classification and data-driven response guidance.

System Architecture Overview



CLIP ViT-L/14
Zero-shot primary

ResNet50 LP
MEDIC fine-tuned

DisasterCNN_v1
Custom trained

RAG + Gemini
Response advisor

Model 1: CLIP ViT-L/14 (Primary Classifier — A2)

Architecture

ViT-L/14 · 431M parameters · Zero-shot

Method

Multi-prompt averaging: 6 classes × 4–7 prompts → cosine similarity mean → ×100 temperature scaling → softmax

Output

Top-3 class probabilities + confidence score (0–1)

Prompt sets

28 total prompts across 6 classes (7 earthquake, 4 flood, 4 fire, 4 typhoon, 5 landslide, 4 other)

6 Disaster Classes

地震或建築損壞 Earthquake Damage

淹水 Flood

火災 Fire

颱風或強風災損 Typhoon / Storm

土石流或坍方 Landslide

其他或無明顯災害 Other / No Disaster

Model 2: ResNet50 Linear Probe (Secondary Classifier — A3)

Architecture

ResNet50 backbone (frozen, ImageNet pretrained)
+ nn.Linear(2048 → 5) replacing fc layer

Training

Fine-tuned on QCRI/MEDIC · 5 classes · 90MB weights
Adam lr=1e-3 · Batch 32 · 10 epochs · WeightedRandomSampler

Val Accuracy

~82–85% (5-class task)

Weights

models/resnet50_linear.pth (90MB)

5-Class Mapping (MEDIC → CrisisLens)

Damaged_Infrastructure → 地震或建築損壞

Fire_Disaster → 火災

Land_Disaster → 土石流或坍方

Non_Damage → 其他或無明顯災害

Water_Disaster → 淹水

Key Design Note: No Typhoon class (intentional). Typhoon images map to Water_Disaster → model_agreement = 0 → need_review = 1. This educationally demonstrates cross-model class-space mismatch.

Dual-Model Comparison: model_agreement & need_review (A4)



need_review = 1 triggered by ANY of:

Confidence < 0.50
CLIP_LOW_CONF_THRESHOLD

Top-2 gap < 0.15
CLIP_TOP2_GAP_THRESHOLD

model_agreement = 0
CLIP ≠ ResNet50 label

All need_review = 1 reports appear in the Admin Dashboard's 'Need Review' filter and the MLOps monitoring page.

Model 3: DisasterCNN_v1 (Custom-Trained CNN)

Architecture

4-block CNN · BatchNorm + AdaptiveAvgPool · ~400K params

Training

QCRI/MEDIC · 6 classes · WeightedRandomSampler
Adam lr=1e-3 · StepLR · 5 epochs · Batch 32

Val Accuracy

68.57% (6-class task)

Weights

models/custom_cnn.pth (1.5 MB)

Role

Standalone auxiliary — 'Custom CNN' mode; CLIP always takes precedence

Architecture

Input (B, 3, 224, 224)

Block 1: Conv(3→32) → BN → ReLU → MaxPool [112×112]

Block 2: Conv(32→64) → BN → ReLU → MaxPool [56×56]

Block 3: Conv(64→128) → BN → ReLU → MaxPool [28×28]

Block 4: Conv(128→256) → BN → ReLU → MaxPool [14×14]

AdaptiveAvgPool(1×1) → Flatten → Dropout(0.3)

Linear(256 → 6) → Softmax

⚠ Typhoon recall ~48% — expected weakness: MEDIC 'hurricane' imagery ≠ Taiwan typhoon visual patterns (domain gap)

B. Citizen Disaster Reporting Flow



What AI does (Step 5)

- CLIP ViT-L/14 classification → Top-3 probabilities
- ResNet50 secondary classification (if mode selected)
- model_agreement computation (label comparison)
- Top-2 gap & confidence threshold checks → need_review
- RAG: FAISS retrieval → Gemini 2.0 Flash advisory generation
- Safety guard: keyword → Gemini safety API → ShieldGemma

What system does (Step 7)

- Rate limit check: max 10 reports/hour/user
- EXIF-stripped image stored (Azure Blob / local)
- Event aggregation v4: disaster type + location + time window
- H3 grid_summary updated (res 9 / district / city fallback)
- model_runs record written: all 7 version strings logged
- Priority Score calculated: $\text{Severity} \times 0.5 + \text{Vuln} \times 0.3 + \text{Cred} \times 0.2$

C. Admin Dashboard & Event Management

Event Dashboard

Priority Score sorting (High/Medium/Low)

Disaster type · city · status filters

Need Review filter checkbox

4 stat cards incl. need_review count

Event status update buttons

Event Detail

Per-report photos + AI predictions

Safety labels (input_safety / output_safety)

Admin Corrections form → admin_corrections table

need_review badges per report

Top-3 confidence breakdown

Priority Score Formula

$\text{Score} = \text{Severity} \times 0.50 + \text{Vulnerability} \times 0.30 + \text{Credibility} \times 0.20$

High ≥ 70 · Medium 40–69 · Low < 40

Severity: injuries, trapped persons, road block, AI confidence

Vulnerability: estimated affected persons

Credibility: report count + model_agreement ratio

Event Status Lifecycle

pending_review → active → resolved → archived

Every transition logged in admin_action_logs
(admin_user · old_value · new_value · reason · timestamp)

Resolved events hidden from H3 heatmap by default

D. H3 Hexagonal Heatmap (GIS)

Multi-Scale Hexagonal Heatmap

H3 Res 5 → 7 → 9 (zoom-based)

Res 5 County level (~100km)

Res 7 Township level (~10km)

Res 9 Neighborhood (~174m)

3-Layer Location Fallback

h3 (GPS) → district → city (no GPS = no heatmap exclusion)

Zoom-Based Resolution Switching

Map zoom level determines which H3 resolution layer is active

Resolved Event Filter

Sidebar toggle: hide/show archived events (default: hide)

Interactive Tooltip

Report count · Priority level · Estimated affected persons

Why Hexagons?

H3 topology: equal-distance neighbors in all 6 directions, no edge/corner bias vs. square grids

E. MLOps Monitoring & Version Tracking

Version Tracking

Every inference writes to
model_runs:

```
clip-vitl14-v1
resnet50-linear-probe-medic-5class-v1
custom-cnn-medic-6class-v1
faiss-multilingual-minilm-v1
disaster-group-distance-timewindow-v4
svcp-weighted-v2
```

Admin Corrections

Human labels in admin_corrections:

- corrected_by · report_id
- original_value → corrected_value
- correction_reason
- used_for_retraining flag (0/1)

MLOps page: pending vs. used counts

Retraining Signals

Trigger conditions:

need_review rate > 30%
over 100 consecutive reports

model_agreement rate < 60%
over any 7-day window

 **need_review > 30% → Retrain!**

F. Data Card (MLSecOps Workshop 1 Format)

QCRI/MEDIC Dataset

~17,600 images (social media)
7 original → 6 CrisisLens classes
80/20 train/val split (seed=42)
WeightedRandomSampler (38% Other)
Bias: Western hemisphere images
Domain gap: hurricane ≠ typhoon
Geographic coverage: 2012–2020
Pre-processing: Resize(256) →
CenterCrop(224) → ImageNet norm

Production Data (CrisisLens)

User-submitted disaster photos
GPS + city/district metadata
EXIF stripped at upload
Key fields:
disaster_type · clip_confidence
need_review · model_agreement
input/output_safety_label
rag_advice (JSON)
Privacy: GPS coords = High risk

RAG Knowledge Base

6 SOP documents (zh-TW)
Taiwan govt disaster guidelines
earthquake / flood / fire
typhoon / landslide / emergency
Chunk: 400 chars / 80 overlap
Embedding: multilingual MiniLM
(384-dimensional vectors)
Index: FAISS Flat L2
Version: faiss-minilm-v1

G. Model Card Summary (MLSecOps Workshop 1 Format)

Aspect	CLIP ViT-L/14	ResNet50 LP	DisasterCNN_v1	RAG System
Role	Primary classifier	Secondary classifier	Auxiliary (standalone)	Response advisor
Training	OpenAI pretrained (zero-shot)	MEDIC fine-tuned 5-class LP	From scratch 6-class full train	Static KB + Gemini LLM
Params	431M	~25M (frozen)	~400K	N/A
Accuracy	Qualitative: High	Val ~82–85%	Val 68.57%	N/A
Failure	Adversarial images	Typhoon→Water (expected)	Typhoon 48% recall	Gemini API unavailable
Mitigation	Confidence + gap thresholds	model_agreement flag → review	CLIP always takes precedence	Static fallback guidelines

H. AI Security & Safety Considerations

1. Adversarial Attacks

Threat: FGSM/PGD pixel perturbations fool CLIP/ResNet50 into classifying real disasters as 'No Disaster'

Defence: Dual-model agreement: attacker must fool BOTH independently-trained models simultaneously. Confidence threshold + Top-2 gap → need_review pipeline.

Ref: *Goodfellow 2015 (FGSM); Madry 2018 (PGD)*

2. RAG Poisoning

Threat: Malicious document injection into FAISS KB → Gemini generates harmful evacuation advice

Defence: Static, version-controlled SOPs (6 files, admin-only rebuild). Index version logged in every model_runs record. Mandatory non-reliance disclaimer on all output.

Ref: *Zou 2024; Greshake 2023*

3. Prompt Injection

Threat: Malicious disaster description text hijacks Gemini 2.0 Flash output via embedded instructions

Defence: 3-layer ShieldGemma guard: keyword filter → Gemini safety API → local ShieldGemma model. User text in <user_report> block, not instruction block. Rate limit: 10 reports/hr.

Ref: *Perez & Ribeiro 2022; Branch et al. 2022*

4. EXIF GPS Leakage

Threat: JPEG EXIF GPS tags expose reporter's precise home/workplace coordinates to unauthorized parties

Defence: strip_exif() in utils/image_utils.py: pixel-level PIL Image rebuild. Zero EXIF in database or Azure Blob. Called unconditionally in load_image() before analysis or storage.

Ref: *GDPR Art. 4(1) — location = personal data*

Platform Demo — Key UI Pages

Citizen Portal

app.py

Upload photo · AI Analyse · Select model (CLIP / ResNet50 / CNN)

GPS location · Disaster conditions · Rate-limited submit

Model Comparison

app.py — sidebar mode

CLIP vs ResNet50 side-by-side

model_agreement badge · need_review badge · Top-2 gap display

Event Dashboard

pages/2_Event_Dashboard.py

Priority-sorted event cards · 4 stat counters

Need Review filter · Disaster type / city / status filters

Event Detail

pages/3_Event_Detail.py

Per-report photos + Top-3 predictions

Safety labels · Admin Corrections form · need_review badges

H3 Heatmap

pages/4_H3_Heatmap.py

Multi-scale hexagonal map over Taiwan

Zoom-based res switching · Resolved event toggle · Tooltips

MLOps Dashboard

pages/6_MLOps.py

3 tabs: Model Runs · Admin Corrections · Need Review

Version tracking · Retraining signals · need_review rate

Conclusion & Future Work

What We Built

- ✓ 3-model NN pipeline: CLIP + ResNet50 + DisasterCNN
- ✓ RAG advisory: FAISS + Gemini 2.0 Flash + fallback
- ✓ Event aggregation v4: type + location + time window
- ✓ H3 multi-scale heatmap (Res 5/7/9 zoom switching)
- ✓ MLOps version tracking (model_runs + admin_corrections)
- ✓ 4 AI security mitigations implemented
- ✓ Data Card + Model Card (MLSecOps Workshop 1 format)
- ✓ EXIF GPS stripping at point of upload
- ✓ Admin Corrections → retraining data pipeline

Future Work

- Formal adversarial evaluation (AutoAttack / certified defence)
- Taiwan-specific fine-tuning dataset collection
- Real-time video stream classification
- Automated retraining CI/CD pipeline
- Multi-label disaster classification
- Federated learning for privacy-preserving fine-tuning
- GitHub Actions CI + automated pytest coverage